

AI Data Center Cooling Risk Decision Report

Risk Findings

Claim: Flow interruption is the highest-severity technical risk because thermal margin can decay before a manual response is effective.

Claim: Sensor drift is a high-likelihood operational risk because it can bias control decisions without an immediate hard failure.

Claim: Shared heat-rejection saturation can affect multiple compute rows at the same time.

Statistic: In the repository-authored stress case, simulated thermal margin falls from 12 minutes to 4 minutes after loss of the active pump path.

Statistic: The synthetic review found 3 of 12 sensors outside the plus or minus 0.5 C calibration tolerance during a quarterly check.

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Response Options and Decision Criteria

Comparison: Option A adds pump-path redundancy and automated failover, giving the strongest interruption protection at the highest implementation effort.

Comparison: Option B increases monitoring and calibration frequency, reducing detection delay at lower effort but leaving pump-path exposure unchanged.

Decision criterion: Prefer an option that protects thermal margin, preserves maintainability, and can be verified through staged commissioning.

Claim: A combined option can address both rapid flow loss and slower sensor degradation without requiring a full plant redesign.

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Recommendation, Implementation, and Limitations

Recommendation: Approve a phased combined response: instrument the baseline, commission automated pump failover, and then tighten sensor calibration governance.

Process: Phase 1 records baseline alarms and flow availability; Phase 2 commissions failover under controlled load; Phase 3 audits sensor drift and operating procedures.

Claim: The staged path creates explicit stop points before each increase in operational dependency.

Limitation: All values in this report are synthetic planning assumptions and require site measurement before an engineering release decision.